

Midterm Exam

Version B

Friday, October 9, 2026 · in class

Name: _____ Student ID: _____

40 points | Suggested working time 40 minutes

Use a basic calculator. Show your working. Allow about 7, 10, 7, 10 and 3 minutes for Sections 1–5, plus 3 minutes to read and check. Sections 1 and 5 are marked on clear reasoning, not on matching one “right” answer. The data are synthetic teaching data from a fictional seaside café.

Section	What it covers	Points	Score
1	Choose the inputs	8	
2	Read the LINEST output	12	
3	Errors and outliers	8	
4	Did the price change sales?	9	
5	Summary for the owner	3	
Total		40	

SECTION 1 · Choose the inputs

8 points · about 7 minutes

A seaside café wants to predict **tomorrow’s iced coffee sales** (cups) **before it opens**. Table 1 lists information it could use. Across 168 days of data, **temperature and sunshine are strongly related to each other ($r = 0.78$)**. Their correlations with iced coffee sales are 0.68 (temperature) and 0.53 (sunshine).

Table 1. Candidate information.

#	Information
1	Tomorrow’s temperature forecast (°C)
2	Tomorrow’s sunshine forecast (hours)
3	Whether tomorrow is a weekend (yes or no)
4	Whether a beach event is on the calendar tomorrow (yes or no)
5	Tomorrow’s planned iced coffee price (€)
6	Tomorrow’s iced coffee revenue (€), calculated after closing

1a Choose **three** inputs for the model. For each, say whether it should push iced coffee sales **up or down**, and give a one-line reason. 3 pts

Input (name or number)	Up or down?	Reason (one line)

1b One candidate in Table 1 is **unsuitable** for this forecast. Which one, and what is the problem? 2 pts

1c Should temperature **and** sunshine both go into the same model? Explain, and say which one you would keep. 2 pts

1d How would you check the model’s prediction errors on days that were **not** used to fit it? 1 pt

SECTION 2 · Read the LINEST output

12 points · about 10 minutes

A model of daily iced coffee sales (cups) was fitted to 168 days with $\text{=LINEST}(y, X, \text{TRUE}, \text{TRUE})$. The three X columns were entered in this order: $X_1 = \text{temperature } (^{\circ}\text{C})$, $X_2 = \text{sunshine (hours)}$, $X_3 = \text{weekend (1 = Saturday or Sunday, 0 = weekday)}$. Use Table 2 to read Table 3.

Table 2. How LINEST lays out its results.

Col A (X_3)	Col B (X_2)	Col C (X_1)	Col D (Intercept)	Statistic description
Slope m_3	Slope m_2	Slope m_1	Intercept (b)	Coefficients for: $y = m_1X_1 + m_2X_2 + m_3X_3 + b$
SE – m_3	SE – m_2	SE – m_1	SE – Intercept	Standard error for each coefficient
R^2	SE – Y	Blank	Blank	Quality of the model fit
F-statistic	df	Blank	Blank	Analysis of variance metrics
SS – Reg	SS – Resid	Blank	Blank	Sum of squares metrics

Table 3. LINEST output for the model (shaded cells are blank).

Col A	Col B	Col C	Col D
17.93	–0.20	4.35	9.78
2.24	0.66	0.48	7.84
0.62	13.11		
88.32	164		
45558.77	28198.94		

2a Write the fitted equation, using the numbers in Table 3 and the variable names.

2 pts

2b Each statement below misreads the output. Rewrite each one correctly, keeping the units and conditions.

3 pts

(i)	"Each extra hour of sunshine reduces iced coffee sales by 0.20 cups, so sunny days are bad for business."
(ii)	"Warm weather makes each customer buy 4.35 more iced coffees for every $^{\circ}\text{C}$."
(iii)	"SE – Y = 13.11 means every forecast is wrong by exactly 13 cups."

2c A coefficient is clearly different from zero if it is **more than 2 standard errors** away from zero. (a) Which one predictor is **not**? Show the comparison. (b) Use Section 1 to suggest why. (c) What would you do with this predictor?

3 pts

2d Tomorrow is forecast to be 24°C with 6 hours of sunshine on a weekend. (a) Predict iced coffee sales. (b) Use SE – Y to give a rough range: prediction $\pm 2 \times \text{SE} - Y$. (c) Actual sales turn out to be **118 cups**. Is that surprising? Explain.

4 pts

SECTION 3 · Errors and outliers

8 points · about 7 minutes

3a · Errors. Sales and errors are in cups. These are constructed exercise numbers, not workbook output. **Residual = actual – predicted.** **MAE = (sum of the absolute residuals) ÷ n.**

Day	Actual	Predicted	Residual
Tue	100	108	
Sun		120	+10
Thu	96		+3

3a-i Fill the blanks in the table.

1 pt

3a-ii Calculate the MAE of these predictions.

1 pt

3a-iii Model Z has residuals of +5, +5, +5 on the same three days. Which model has the lower MAE? What do the identical signs tell you about Model Z?

2 pts

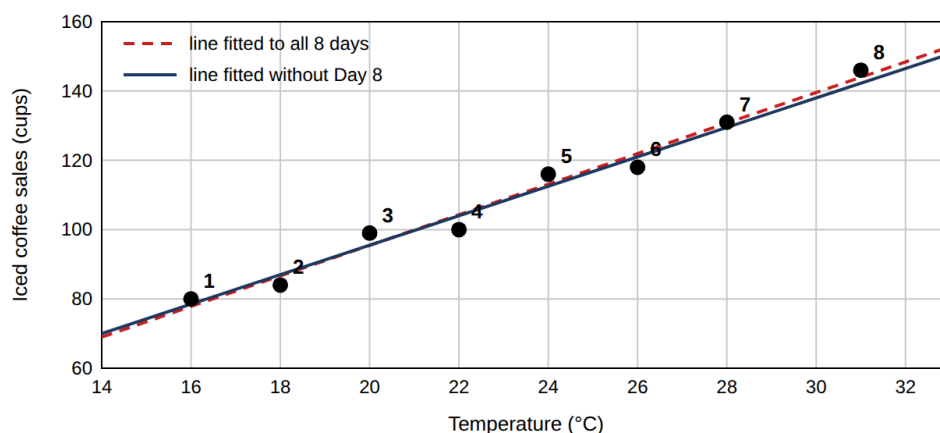
3b · Outliers. Table 4 and Figure 1 show iced coffee sales on 8 days (constructed example). Table 5 gives the straight-line results with and without Day 8.

Day	1	2	3	4	5	6	7	8
Temperature (°C)	16	18	20	22	24	26	28	31
Sales (cups)	80	84	99	100	116	118	131	146

Table 5. Results for the straight line (sales against temperature).

	All 8 days	Without Day 8
Slope (cups per °C)	4.41	4.25
R ²	0.98	0.97
SE – Y (cups)	3.43	3.55

Figure 1. The 8 days, with the line fitted to all of them (dashed) and the line fitted without Day 8 (solid).



A classmate says: “Delete Day 8, it is an outlier.”

3b-i Does Day 8 follow the pattern of the other days? How can you tell from Figure 1?

1 pt

3b-ii What impact does Day 8 have on the slope and on R²?

2 pts

3b-iii Would you delete it? Give a reason.

1 pt

SECTION 4 · Did the price change sales?

9 points · about 10 minutes

On 30 June the café cut the iced coffee price from **€3.50** to **€3.10**. Table 6 shows iced coffee sales (cups) on 5 weekdays **before** (Group 1) and 5 weekdays **after** (Group 2). The average temperature was 21 °C in the “before” sample and 25 °C in the “after” sample.

Formulas. $s = \sqrt{[\sum (x - \bar{x})^2 \div (n - 1)]}$ · $s_p = \sqrt{[(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] \div (n_1 + n_2 - 2)}$ · $t = (\bar{x}_1 - \bar{x}_2) \div [s_p \times \sqrt{(1/n_1 + 1/n_2)}]$ · $df = n_1 + n_2 - 2$

For $df = 8$ the two-tailed 5% critical value is 2.31. A negative t only means that Group 2 has the larger mean.

Table 6. Worksheet (fill in the blank cells).

Day	Before (x_1)	$x_1 - \bar{x}_1$	$(x_1 - \bar{x}_1)^2$	After (x_2)	$x_2 - \bar{x}_2$	$(x_2 - \bar{x}_2)^2$
1	100			110		
2	108			118		
3	92			102		
4	104			114		
5	96			126		
Sum						
Mean $\bar{x}$						
s						

4a Complete Table 6: the two means, the two sums of squared deviations, and s for each group (2 decimals). **3 pts**

4b Calculate s_p and t (2 decimals). **2 pts**

Sheets gives **=T.TEST(B2:B6, C2:C6, 2, 2) = 0.021** (two-tailed, equal variances).

4c (i) State the null hypothesis in words. (ii) Is the difference statistically significant at 5%? Explain using the p -value and your t . (iii) In plain words, what does $p = 0.021$ mean? **3 pts**

4d The “after” days were warmer. What does that do to a conclusion that the price cut caused the rise? **1 pt**

SECTION 5 · Summary for the owner

3 points · about 3 minutes

5 Write 3–4 sentences to the café owner, who has no statistics background. Include: (i) which inputs you recommend for forecasting iced coffee sales and the main finding from the LINEST output; (ii) what the price test showed; (iii) one caution. **3 pts**

— End of exam —